

8 Probability Rules

8.1 Multiplication rule

- Rearranging the definition of conditional probability we get the **Multiplication rule**: the probability that two events both occur is

$$\begin{aligned}P(A \cap B) &= P(A|B)P(B) \\ &= P(B|A)P(A)\end{aligned}$$

- The multiplication rule says that you should think “multiply” when you see “and”.
- However, be careful about *what* you are multiplying: to find a joint probability you need an unconditional and an appropriate conditional probability.
- You can condition either on A or on B , provided you have the appropriate marginal probability; often, conditioning one way is easier than the other.
- Be careful: the multiplication rule does *not* say that $P(A \cap B)$ is the same as $P(A)P(B)$.
- The multiplication rule is useful in situations where conditional probabilities are easier to obtain directly than joint probabilities.

Example 8.1. A standard deck of playing cards has 52 cards, 13 cards (2 through 10, jack, king, queen, ace) in each of 4 suits (hearts, diamonds, clubs, spades). Shuffle a deck and deal cards one at a time without replacement.

1. Find the probability that the first card dealt is a heart.
2. If the first card dealt is a heart, determine the conditional probability that the second card is a heart.

3. Find the probability that the first two cards dealt are hearts.

4. Find the probability that the first two cards dealt are hearts and the third card dealt is a diamond.

- The multiplication rule extends naturally to more than two events (though the notation gets messy). For three events, we have

$$P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2)$$

- And in general,

$$P(A_1 \cap A_2 \cap A_3 \cap A_4 \cap \dots) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2)P(A_4|A_1 \cap A_2 \cap A_3) \dots$$

- The multiplication rule is useful for computing probabilities of events that can be broken down into component “stages” where conditional probabilities at each stage are readily available. At each stage, condition on the information about all previous stages.

Example 8.2. The [birthday problem](#) concerns the probability that at least two people in a group of n people have the same birthday¹. Ignore multiple births and February 29 and assume that the other 365 days are all equally likely².

1. If $n = 30$, what do you think the probability that at least two people share a birthday is: 0-20%, 20-40%, 40-60%, 60-80%, 80-100%? How large do you think n needs to be in order for the probability that at least two people share a birthday to be larger than 0.5? Just make guesses before proceeding to calculations.

¹You should really click on [this birthday problem link](#).

²Which isn't [quite true](#). However, a non-uniform distribution of birthdays only increases the probability that at least two people have the same birthday. To see that, think of an extreme case like if everyone were born in September.

2. Now consider $n = 3$ people, labeled 1, 2, and 3. What is the probability that persons 1 and 2 have different birthdays?
3. What is the probability that persons 1, 2, and 3 all have different birthdays *given* that persons 1 and 2 have different birthdays?
4. What is the probability that persons 1, 2, and 3 all have different birthdays?
5. When $n = 3$. What is the probability that at least two people share a birthday?
6. For $n = 30$, find the probability that at least two people have the same birthday.
7. Write a clearly worded sentence interpreting the probability in the previous part as a long run relative frequency.
8. When $n = 100$ the probability is about 0.9999997. If you are in a group of 100 people and no one shares your birthday, should you be surprised? Discuss.

8.2 Law of total probability

Example 8.3. There exist multiple penguin species throughout Antarctica, including the *Adelie* (A), *Chinstrap* (C), and *Gentoo* (G). Suppose that among Antarctic penguins³ of these three species, 44.2% of Adelie, 19.8% are Chinstrap, and 36.0% are Gentoo. Suppose that 83.4% of Adelie penguins are below average weight, 89.7% of Chinstrap penguins are below average weight, and 4.9% of Gentoo penguins are below average weight (that is, below 4200g).

Randomly select a penguin. Compute the probability that it is below average weight.

- **Law of total probability.** If $C_1, \dots, C_k$ are disjoint with $C_1 \cup \dots \cup C_k = \Omega$, then

$$\begin{aligned} P(A) &= \sum_{i=1}^k P(A \cap C_i) \\ &= \sum_{i=1}^k P(A|C_i)P(C_i) \end{aligned}$$

- The events $C_1, \dots, C_k$, which represent the “cases”, form a *partition* of the sample space; each outcome $\omega \in \Omega$ lies in exactly one of the C_i .
- The law of total probability says that we can interpret the unconditional probability $P(A)$ as a probability-weighted average of the case-by-case conditional probabilities $P(A|C_i)$ where the weights $P(C_i)$ represent the probability of encountering each case.
- Conditioning and using the law of probability is an effective strategy in solving many problems, even when the problem doesn’t seem to involve conditioning.
- For example, when a problem involves iterations or steps it is often useful to *condition on the result of the first step*.

Example 8.4. You and your friend are playing the “lookaway challenge”.

The game consists of possibly multiple rounds. In the first round, you point in one of four directions: up, down, left or right. At the exact same time, your friend also looks in one of those four directions. If your friend looks in the same direction you’re pointing, you win! Otherwise, you switch roles and the game continues to the next round — now your friend points in a direction and you try to look away. As long as no one wins, you keep switching

³Throughout “penguin” refers to an Antarctic penguin of one of these three species. This example is based on the famous [Palmer penguins data set](#).

off who points and who looks. The game ends, and the current “pointer” wins, whenever the “looker” looks in the same direction as the pointer.

Suppose that each player is equally likely to point/look in each of the four directions, independently from round to round. What is the probability that you win the game?

1. Why might you expect the probability to not be equal to 0.5?
2. If you start as the pointer, what is the probability that you win in the first round?
3. If p denotes the probability that the player who starts as the pointer wins the game, what is the probability that the player who starts as the looker wins the game? (Note: p is the probability that the person who starts as pointer wins the whole game, not just the first round.)
4. Let A be the event that the person who starts as the pointer wins the game, and B be the event that the person who starts as the pointer wins in the first round. What is $P(A|B)$?
5. Find a simple expression for $P(A|B^c)$ in terms of p . The key is to consider this question: if the player who starts as the pointer does not win in the first round, how does the game behave from that point forward?

6. Condition on the result of the first round and set up an equation to solve for p .

7. Interpret the probability from the previous part.

8.3 Bayes Rule

- **Bayes' rule for events** specifies how a prior probability $P(H)$ of event H is updated in response to the evidence E to obtain the posterior probability $P(H|E)$.

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)}$$

- Event H represents a particular hypothesis (or model or case)
- Event E represents observed evidence (or data or information)
- $P(H)$ is the unconditional or **prior probability** of H (prior to observing evidence E)
- $P(H|E)$ is the conditional or **posterior probability** of H after observing evidence E .
- $P(E|H)$ is the **likelihood** of evidence E given hypothesis (or model or case) H
- Bayes rule is often used when there are multiple hypotheses or cases. Suppose $H_1, \dots, H_k$ is a series of distinct hypotheses which together account for all possibilities, and E is any event (evidence).
- Combining Bayes' rule with the law of total probability,

$$\begin{aligned} P(H_j|E) &= \frac{P(E|H_j)P(H_j)}{P(E)} \\ &= \frac{P(E|H_j)P(H_j)}{\sum_{i=1}^k P(E|H_i)P(H_i)} \end{aligned}$$

Example 8.5. Continuing Example 8.3. Suppose that among Antarctic penguins of these three species, 44.2% of Adelie, 19.8% are Chinstrap, and 36.0% are Gentoo. Suppose that 83.4% of Adelie penguins are below average weight, 89.7% of Chinstrap penguins are below average weight, and 4.9% of Gentoo penguins are below average weight (that is, below 4200g).

Randomly select a penguin and suppose it is below average weight. We are interested in classifying the species of the penguin.

1. Frame this problem in the context of Bayes rule: Identify the hypotheses, prior probabilities, evidence, likelihood, and posterior probabilities.
 2. Prior to observing the penguin's weight, how many times more likely is it to be Adelie than to be Gentoo?
 3. How many times more likely is it for a penguin to be below average weight given that it is Adelie than given that it is Gentoo?
 4. Compute the posterior probability of each species given that the penguin is below average weight. Given that the penguin is below average weight, how many times more likely is it to be Adelie than to be Gentoo?
 5. How are the ratios in the three previous parts related?
- Bayes rule is often used when there are multiple hypotheses or cases. Suppose $H_1, \dots, H_k$ is a series of distinct hypotheses which together account for all possibilities, and E is any

event (evidence).

$$\begin{aligned} P(H_j|E) &= \frac{P(E|H_j)P(H_j)}{P(E)} \\ &= \frac{P(E|H_j)P(H_j)}{\sum_{i=1}^k P(E|H_i)P(H_i)} \end{aligned}$$

$$P(H_j|E) \propto P(E|H_j)P(H_j)$$

- The symbol $\propto$ is read “is proportional to”. The relative *ratios* of the posterior probabilities of different hypotheses are determined by the product of the prior probabilities and the likelihoods, $P(E|H_j)P(H_j)$.
- The marginal probability of the evidence, $P(E)$, in the denominator simply normalizes the numerators to ensure that the updated conditional probabilities given the evidence sum to 1 over all the distinct hypotheses.
- **In short, Bayes’ rule says that posterior is proportional to the product of prior and likelihood**

$$\text{posterior} \propto \text{prior} \times \text{likelihood}$$

Example 8.6. Continuing Example 8.5. Suppose that among Antarctic penguins of these three species, 44.2% of Adelie, 19.8% are Chinstrap, and 36.0% are Gentoo. Suppose that 83.4% of Adelie penguins are below average weight, 89.7% of Chinstrap penguins are below average weight, and 4.9% of Gentoo penguins are below average weight (that is, below 4200g).

Randomly select a penguin. We are interested in classifying the species of the penguin.

1. Before you know the weight of the penguin (or any other information) what specify would you predict it to be? Why? What is the probability that your classification is correct?
2. If the penguin is below average weight, what species would you classify it as? Why?

3. Now suppose that the penguin is not below average weight. Use a Bayes table to compute the posterior probability of each species given that the penguin is not below average weight. If the penguin is not below average weight, what species would you classify it as? Why?

4. Suppose that you classify any randomly selected penguin based on whether or not it is below average weight as in the two previous parts. What is the posterior probability that you classify the penguin correctly? How does this compare to your probability of being correct before you observed the penguin's weight?

- Bayesian analysis is often an iterative process.
- Posterior probabilities are updated after observing some information or data. These probabilities can then be used as prior probabilities before observing new data.
- Posterior probabilities can be sequentially updated as new data becomes available, with the posterior probabilities after the previous stage serving as the prior probabilities for the next stage.
- The final posterior probabilities only depend upon the cumulative data. It doesn't matter if we sequentially update the posterior after each new piece of data or only once after all the data is available; the final posterior probabilities will be the same either way. Also, the final posterior probabilities are not impacted by the order in which the data are observed.

Example 8.7. Suppose that you are presented with six boxes, labeled 0, 1, 2, ..., 5, each containing five marbles. Box 0 contains 0 green and 5 gold marbles, box 1 contains 1 green and 4 gold, and so on with box i containing i green and $5 - i$ gold. One of the boxes is chosen uniformly at random (perhaps by rolling a fair six-sided die), and then you will randomly select marbles from that box, without replacement. Imagine the boxes appear identical and you can't see inside; all you observe is the color of the marbles you select. Based on the colors of the marbles selected, you will update the probabilities of which box had been chosen.

1. Suppose that a single marble is selected and it is green. Which box do you think is the most likely to have been chosen? Make a guess for the posterior probabilities for each box. Then construct a Bayes table to compute the posterior probabilities. How do they compare to the prior probabilities?

2. Now suppose a second marble is selected from the same box, without replacement, and its color is gold. Which box do you think is the most likely to have been chosen given these two marbles? Make a guess for the posterior probabilities for each box. Then construct a Bayes table to compute the posterior probabilities, *using the posterior probabilities from the previous part after the selection of the green marble as the new prior probabilities before seeing the gold marble.*

3. Now construct a Bayes table corresponding to the original prior probabilities ($1/6$ each) and the combined evidence that the first ball selected was green and the second was gold. How do the posterior probabilities compare to the previous part?